DSAA3073: Theories in Data Science

Tutorial 3A: Compete with the Best in Class

150-minute core tutorial · 15-minute break · 60-minute protected buffer

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Explorative projection

LESSON BACKBONE

When no classifier in the declared class is perfect, what lets a data-selected ERM rule compete with the best member of that class, and when is such learning possible at all?

Chapter 2 gave us empirical risk minimization, VC dimension, and the Sauer-Shelah bound. Those results answer an important realizable question, but they do not yet describe noisy data, where even the best classifier in $\mathcal{H}$ makes mistakes.

We will replace perfection by a best-in-class benchmark, identify the one simultaneous sample event that protects ERM, and derive the complete excess-risk comparison. We then place that argument inside the Fundamental Theorem of Statistical Learning and expose why an unrestricted class must fail on unseen labels. One difficult implication remains deliberately open: proving that finite VC dimension supplies the needed simultaneous event.

The three core blocks take **45, 55, and 50 minutes**. Take the scheduled 15-minute break after the first block. The final 60 minutes are protected for questions, slower reconstruction, and recovery; they introduce no required knowledge.

Clock	Mathematical work	Protected time
0-45	Replace realizability by an agnostic best-in-class contract	45 min core
45-60	Scheduled break	15 min
60-115	Define uniform convergence and derive the ERM excess-risk chain	55 min core
115-165	State the scoped equivalence and reconstruct the No-Free-Lunch mechanism	50 min core
165-225	Questions, recovery, alternative calculations	60 min buffer

When zero risk is not available

Chapter 2's realizable model assumed that some $h^\star \in \mathcal{H}$ has $R(h^\star) = 0$. Keep the same binary prediction setting and zero-one loss, but now let the data come from a joint distribution D on $\mathcal{X} \times \{0,1\}$. The same feature vector may receive different labels on different draws. Population risk is therefore

$$R(h) = \mathbb{P}_{(X,Y)\sim D}(h(X) \neq Y), \tag{1}$$

and the training sample is $S = ((x_i, y_i))_{i=1}^m \sim D^m$. We continue to write

$$\hat{R}_{S(h)} = \frac{1}{m}\sum_{i=1}^{m} \mathbf{1}_{\{h(x_i)\neq y_i\}} \tag{2}$$

for empirical risk. These are the same registered risk symbols as in Chapter 2; only the realizability assumption has been removed.

Consider a class of three diagnostic rules. Their population risks are not known to the learner, but suppose an audit later reveals the following values.

rule	$R(h)$	$\widehat{R}_{S(h)}$	interpretation
h_1	0.15	0.16	best population risk in the class
h_2	0.18	0.13	empirical minimizer on this sample
h_3	0.23	0.22	worse by both measures

Table 1: Noise leaves a positive best-in-class baseline. The random sample can still make another rule look empirically better.

If the accuracy tolerance is $\varepsilon = 0.04$, demanding $R(A(S)) \leq \varepsilon$ is impossible even for the best member of this class. The meaningful target is instead

$$R(A(S)) \leq \inf_{h\in\mathcal{H}} R(h) + \varepsilon = 0.19. \tag{3}$$

The number ε measures allowed **excess risk** above the best achievable population risk in the declared class. It is not distance from zero.

Definition: Agnostic PAC learnability

A binary hypothesis class $\mathcal{H}$ is **agnostic PAC learnable** if there are a learning algorithm A and a sample-complexity function $m_{\mathcal{H}} : (0,1)^2 \to \mathbb{N}$ such that, for every $\varepsilon, \delta \in (0,1)$, every joint distribution D on $\mathcal{X} \times \{0,1\}$, and every $m \geq m_{\mathcal{H}}(\varepsilon, \delta)$, an i.i.d. sample $S \sim D^m$ satisfies

$$\mathbb{P}\Big(R(A(S)) \leq \inf_{h\in\mathcal{H}} R(h) + \varepsilon\Big) \geq 1 - \delta. \tag{4}$$

The probability is over the random sample and over any internal randomness used by A.

Read the quantifiers in order. One algorithm must work for every distribution; the required sample size may depend on $\mathcal{H}$, ε, and δ, but not on the unknown D. The output is judged against the best risk available inside $\mathcal{H}$, even if a better predictor exists outside it.

The infimum is deliberate. A class need not contain a hypothesis attaining its smallest possible risk. For any comparison tolerance $\rho > 0$, the definition of infimum guarantees a comparator $h_\rho \in \mathcal{H}$ with

$$R(h_\rho) \le \inf_{h\in\mathcal{H}} R(h) + \rho. \tag{5}$$

We may prove a bound against h_ρ and then let ρ approach zero. This avoids silently assuming that a population minimizer exists.

IN-CLASS CHECK

Prompt: A diagnostic class has best-in-class population risk 0.12. For $\varepsilon = 0.03$ and $\delta = 0.05$, which guarantee is agnostic PAC: (a) output risk at most 0.03 on every sample; (b) output risk at most 0.15 with probability at least 0.95; or (c) expected output risk at most 0.15? Explain each rejected choice.

Hint: Attach ε to the best-in-class baseline and attach δ to a high-probability event.

Answer:

> Choice (b) has the required form: $0.12 + 0.03 = 0.15$ and success probability at least $1 - 0.05 = 0.95$. Choice (a) incorrectly treats ε as an absolute risk target and is below the class optimum. Choice (c) replaces a high-probability statement by an expectation statement; the latter does not by itself supply the declared confidence guarantee.

IN-CLASS CHECK

Prompt: Reconstruct the agnostic PAC statement from these cards: one learner; every joint distribution; every (ε, δ); sufficiently large i.i.d. sample; probability at least $1-\delta$; best-in-class risk plus ε. Put them in logical order and identify what is random.

Hint: Begin with the existence of a learner and sample-complexity function.

Answer:

> There exist A and $m_{\mathcal{H}}$ such that, for every $\varepsilon, \delta \in (0,1)$ and every joint distribution D, whenever $m \ge m_{\mathcal{H}}(\varepsilon, \delta)$ and $S \sim D^m$, the random output obeys $R(A(S)) \le \inf_{h\in\mathcal{H}} R(h) + \varepsilon$ with probability at least $1 - \delta$. Randomness comes from S and, if the learner is randomized, from its internal random choices.

Two different approximations

An implementation may also return only an approximate empirical minimizer. We say that $h_S = A(S)$ is an η-approximate ERM output when

$$\hat{R}_{S(h_S)} \le \inf_{h\in\mathcal{H}} \hat{R}_{S(h)} + \eta, \tag{6}$$

where $\eta \geq 0$ is an optimization tolerance. Do not merge η with the comparison tolerance ρ: η concerns how well the algorithm solves the empirical optimization problem; ρ is a proof device used when the population infimum is unattained.

Worked example: Decode the diagnostic benchmark

In the table, h_2 is an exact ERM because its empirical risk 0.13 is the smallest displayed value. Its population excess risk is $0.18 - 0.15 = 0.03$. Thus it meets an agnostic target with $\varepsilon = 0.04$ on this particular sample. This one successful sample is not a PAC proof: a PAC statement must control the probability of success over fresh samples from every allowed distribution.

IN-CLASS CHECK

Prompt: Suppose an optimizer returns h_S with empirical risk 0.006 above the empirical infimum. A proof budget allows total excess $\varepsilon = 0.04$. If two population-to-empirical comparisons each cost $\alpha = 0.015$, does the proof close? What quantity remains unused?

Hint: Add two deviation tolerances and one optimization tolerance.

Answer:

Yes. The accounted excess is $2\alpha + \eta = 2(0.015) + 0.006 = 0.036 \leq 0.04$. The unused margin is 0.004. A separate comparison tolerance ρ may be taken arbitrarily small when the population infimum is not attained; after proving the inequality for every $\rho > 0$, we let ρ approach zero.

The benchmark is now clear, but the diagnostic table exposes the obstacle: ERM chooses a row after seeing the same random empirical risks we want to trust. A fixed-hypothesis concentration bound does not automatically protect that selected row. We need one event that holds for every row at once.

One event for every possible ERM choice

For a fixed h, concentration can make $\hat{R}_{S(h)}$ close to $R(h)$. Yet the identity of h_S depends on S. The right event must therefore be true before we know which hypothesis ERM will select.

Definition: Uniform convergence

A hypothesis class $\mathcal{H}$ has the **uniform convergence property** for binary zero-one loss if, for every $\alpha, \delta \in (0,1)$, there is a finite $m_{\mathcal{H}}^{\mathrm{UC}}(\alpha, \delta)$ such that every m at least this large obeys

$$\mathbb{P}_{S\sim D^m}\left(\sup_{h\in\mathcal{H}}\left|\hat{R}_{S(h)} - R(h)\right| \le \alpha\right) \ge 1-\delta \tag{7}$$

for every joint distribution D on $\mathcal{X} \times \{0,1\}$.

Call the event inside the probability E_α. On E_α, the same sample satisfies

$$\left|\hat{R}_{S(h)} - R(h)\right| \le \alpha \quad \text{for every } h \in \mathcal{H}. \tag{8}$$

This includes the random h_S and any fixed or near-best population comparator. A pointwise statement such as $\mathbb{P}\left(\left|\hat{R}_{S(h)} - R(h)\right| \le \alpha\right) \ge 1-\delta$ for each fixed h does not say that all hypotheses are simultaneously good on one shared sample.

hypothesis	$R(h)$	$\hat{R}_{S(h)}$	covered by E_α?
h_S	unknown at selection time	empirically minimal	yes, because the event is simultaneous
h_ρ	within ρ of the population infimum	random sample value	yes, because it belongs to $\mathcal{H}$
any other h	population risk	empirical risk	yes, by the same supremum event

Table 2: The uniform event is chosen to cover both rows needed in the proof, including the data-selected row.

IN-CLASS CHECK

Prompt: A sample satisfies the deviation bound for every hypothesis except the row ERM selects. Can that sample belong to E_α? Would separate high-probability statements for each fixed row repair the problem when $\mathcal{H}$ is infinite?

Hint: Read the supremum literally, then distinguish a shared event from many separate events.

Answer:

No. If the selected row violates the bound, the supremum over $\mathcal{H}$ exceeds α, so E_α fails. Separate fixed-row statements do not by themselves produce one simultaneous event for an infinite class. Additional complexity control is needed to prove that the supremum is small.

Follow the selected row across the two risks

Let h_S be an η-approximate ERM, and let h_ρ satisfy $R(h_\rho) \le \inf_{h\in\mathcal{H}} R(h) + \rho$. On E_α, every step below is now licensed:

$R(h_S)$	$\le$	$\hat{R}_{S(h_S)} + \alpha$	$\le$	$\hat{R}_{S(h_\rho)} + \alpha + \eta$	$\le$	$R(h_\rho) + 2\alpha + \eta$
	uniform event on h_S		approximate ERM comparison		uniform event on h_ρ	

Table 3: Two deviation moves and one optimization move account for the complete excess tolerance. Every inequality has a visible reason.

Written as a derivation,

$$
\begin{aligned}
R(h_S) &\le \hat{R}_{S(h_S)} + \alpha \\
&\le \inf_{h\in\mathcal{H}} \hat{R}_{S(h)} + \eta + \alpha \\
&\le \hat{R}_{S(h_\rho)} + \eta + \alpha \\
&\le R(h_\rho) + 2\alpha + \eta \\
&\le \inf_{h\in\mathcal{H}} R(h) + \rho + 2\alpha + \eta.
\end{aligned} \tag{9}
$$

Because this holds for every $\rho > 0$, let $\rho \to 0$.

Result: Agnostic ERM excess-risk guarantee

On the event E_α, every η-approximate ERM output satisfies

$$R(h_S) - \inf_{h\in\mathcal{H}} R(h) \le 2\alpha + \eta. \tag{10}$$

Therefore, if uniform convergence holds with $2\alpha + \eta \le \varepsilon$, approximate ERM is an agnostic PAC learner with accuracy ε and the same failure probability δ.

For exact ERM, $\eta = 0$, so the familiar choice is $\alpha = \frac{\varepsilon}{2}$. The factor two is not a mysterious theorem constant: it records the two times the proof crosses between population and empirical risk.

IN-CLASS CHECK

Prompt: Fill the reason for each move: $R(h_S)$ to $\hat{R}_{S(h_S)}$; $\hat{R}_{S(h_S)}$ to $\hat{R}_{S(h_\rho)}$; and $\hat{R}_{S(h_\rho)}$ to $R(h_\rho)$. Then explain why a bound holding only for the fixed comparator is insufficient.

Hint: The first and third moves use the same event; the middle move uses the optimization promise.

Answer:

> The first move uses E_α on the selected h_S and costs α. The middle move uses η-approximate empirical minimization and costs η. The third move uses E_α on h_ρ and costs another α. A fixed-comparator bound controls only the third move; it leaves the data-selected h_S unprotected, so the proof cannot even begin.

IN-CLASS CHECK

Prompt: A proof establishes E_α with $\alpha = 0.012$ and confidence 0.99. The optimizer has $\eta = 0.005$. State the resulting excess-risk and failure-probability guarantees. Would this meet $(\varepsilon, \delta) = (0.03, 0.01)$?

Hint: Compute $2\alpha + \eta$ and preserve the same event probability.

Answer:

> The excess risk is at most $2(0.012) + 0.005 = 0.029$ on an event of probability at least 0.99, so the failure probability is at most 0.01. Since $0.029 \le 0.03$, the guarantee meets the requested pair.

Check the scale on a finite class

For each fixed h under zero-one loss, Hoeffding's inequality gives

$$\mathbb{P}\left(\left|\hat{R}_{S(h)} - R(h)\right| > \alpha\right) \le 2\exp(-2m\alpha^2). \tag{11}$$

When $\mathcal{H}$ is finite, a union bound over the class yields

$$\mathbb{P}(E_\alpha^c) \le 2|\mathcal{H}|\exp(-2m\alpha^2). \tag{12}$$

Thus it is sufficient that

$$m \ge \frac{\ln\left(2\frac{|\mathcal{H}|}{\delta}\right)}{2\alpha^2}. \tag{13}$$

For exact ERM, substitute $\alpha = \frac{\varepsilon}{2}$:

$$m \ge 2\frac{\ln\left(2\frac{|\mathcal{H}|}{\delta}\right)}{\varepsilon^2}. \tag{14}$$

This recovers logarithmic dependence on class size and the agnostic $\frac{1}{\varepsilon^2}$ scale. Do not merge it with Chapter 2's consistent realizable route, whose leading accuracy scale is $\frac{1}{\varepsilon}$. The assumptions and bad events in the two proofs are different.

Worked example: A finite diagnostic library

Let $|\mathcal{H}| = 100$, $\varepsilon = 0.10$, and $\delta = 0.05$. Exact ERM with the finite-class calculation requires, as a sufficient bound,

$$m \geq 2\frac{\ln\left(2\frac{100}{0.05}\right)}{(0.10)^2} = 200\ln(4000) \approx 1658.81. \tag{15}$$

Hence the smallest integer satisfying this displayed sufficient inequality is 1659. The calculation certifies a worst-case distribution-free bound; it is not a prediction of typical empirical performance.

IN-CLASS CHECK

Prompt: If the finite class grows from 100 to $10,000$ hypotheses while ε and δ stay fixed, by what additive amount does the sufficient sample bound change? Express the answer exactly before approximating.

Hint: Subtract the two logarithms; the factor 100 enters inside a logarithm.

Answer:

The increase is

$$\frac{2}{\varepsilon^2}\ln\left(\frac{10,000}{100}\right) = \frac{2}{\varepsilon^2}\ln(100). \tag{16}$$

For $\varepsilon = 0.10$, the continuous bound rises by $200\ln(100) \approx 921.03$. The displayed integer certificates must be compared at their endpoints: they rise from 1659 to 2580, so their difference is exactly $2580 - 1659 = 921$ samples. The dependence is logarithmic in class size, not linear.

For an infinite class, $|\mathcal{H}|$ no longer closes the union bound. Chapter 2 suggests the right combinatorial quantity: finite VC dimension limits the number of label patterns on finite sets. The fact that this is exactly the boundary between learnability and impossibility is the content of the next theorem; the proof of its difficult simultaneous-convergence arrow is not yet available.

The complete boundary of binary learnability

The theorem below lives inside a precise frame. The tasks are binary classification, the loss is zero-one loss, samples are i.i.d., and guarantees are distribution-free. We also assume suitable measurability: the risks, suprema, and events whose probabilities we write must be measurable. This qualification prevents formal probability statements from being applied to pathological classes; we do not use it as an extra proof technique.

Definition: Fundamental Theorem of Statistical Learning

In distribution-free binary classification with zero-one loss, under suitable measurability conditions, the following properties of a hypothesis class $\mathcal{H}$ are equivalent:

1. $\mathcal{H}$ has the uniform convergence property;
2. every ERM rule is a successful agnostic PAC learner;
3. $\mathcal{H}$ is agnostic PAC learnable;
4. $\mathcal{H}$ is realizably PAC learnable;
5. every ERM rule is a successful realizable PAC learner;
6. $\mathrm{VCdim}(\mathcal{H}) < \infty$.

The statement is not an unqualified equivalence for arbitrary losses or arbitrary nonmeasurable function classes. Within its frame, finite VC dimension is both sufficient and necessary for learnability.

SCOPE FRAME - CIRCLE THESE CONDITIONS

binary classification / zero-one loss	distribution-free / i.i.d. samples	suitable measurability

finite VC dimension	- - - ->	uniform convergence	→	agnostic ERM succeeds

OPEN: Sauer-Shelah is known input; finite VC to uniform convergence is not proved here. Tutorial 3B closes this gap.

agnostic learnability	→	realizable learnability
realizable learnability	→	finite VC dimension

Table 4: Solid arrows use mechanisms already available: the ERM comparison, dropping to the realizable case, and the No-Free-Lunch contrapositive. The orange dashed/open finite-VC-to-uniform connector is a future proof obligation.

IN-CLASS CHECK

Prompt: Place these cards on the map: the ERM excess-risk chain; "agnostic implies realizable"; Sauer-Shelah; and the No-Free-Lunch contrapositive. Which card is available input for a still-open arrow rather than a complete proof of that arrow? Circle every assumption in the theorem frame.

Hint: The combinatorial bound does not by itself compare empirical and population risk.

Answer:

The ERM chain proves uniform convergence implies agnostic ERM success. Dropping to distributions with deterministic realizable labels proves agnostic learnability implies realizable learnability. The No-Free-Lunch contrapositive proves realizable learnability requires finite VC dimension. Sauer-Shelah is available combinatorial input for the finite-VC-to-uniform arrow, but it does not alone complete the statistical proof. The frame assumptions are binary labels, zero-one loss, distribution-free i.i.d. sampling, and suitable measurability.

Why unseen labels defeat an unrestricted learner

Definition: No-Free-Lunch theorem

For every learning algorithm A for binary classification with zero-one loss and every finite sample size m with $2m \leq |\mathcal{X}|$, there is a realizable distribution D for which, with a nontrivial constant probability over the sample and any learner randomness, $A(S)$ has nontrivial population error. In the standard quantitative statement, one may choose the problem so that

$$\mathbb{P}\left(R(A(S)) \geq \frac{1}{8}\right) \geq \frac{1}{7} \tag{17}$$

even though some target classifier has risk zero.

The quantifier order matters: **for every learner, there exists a hard problem**. The theorem does not say that every learner fails on every distribution.

Here is the necessity mechanism inside a class of infinite VC dimension. Fix the learner and its sample size m. Infinite VC dimension supplies a set

$$C = \{x_1, \ldots, x_{2m}\} \tag{18}$$

that $\mathcal{H}$ shatters. Consequently, every binary labeling of C is realized by some $h \in \mathcal{H}$. Draw examples uniformly from C and label them by one of these target functions.

The sample contains at most m distinct points, so at least m points of C remain unseen. Their total probability mass is at least $\frac{1}{2}$. If the target labeling is chosen uniformly from all labelings that the shattered set supports, then after observing the sample each unseen label is still an independent fair bit from the learner's point of view. Whatever label the learner predicts there, it is wrong with conditional probability $\frac{1}{2}$.

x_1 seen label 1	x_2 seen label 0	x_3 seen label 1	x_4 unseen label ?	x_5 unseen label ?	x_6 unseen label ?

With $2m = 6$ and only $m = 3$ observations, at least half the support is unseen. Shattering lets the hidden labels vary without changing the observed labeled sample.

Table 5: Observed agreement does not determine the labels on unseen shattered points. The uncertainty is about information, not computation.

The two halves multiply: at least $\frac{1}{2}$ of the mass is unseen, and on average the learner misses $\frac{1}{2}$ of that mass. This produces constant expected error for some realizable target. Averaging over target labelings then identifies one fixed hard target; a bounded-variable argument converts the expectation into a constant-probability failure. The complete averaging and constant calculation are collected in the zero-minute appendix.

IN-CLASS CHECK

Prompt: In the six-point picture, the learner saw the first three labels and predicts $(0, 0, 1)$ on the three unseen points. Give two target labelings that agree on all observed points but make different unseen predictions fail. Why must both targets be realizable by $\mathcal{H}$?

Hint: Keep the observed prefix $(1, 0, 1)$ fixed and vary the final three bits.

Answer:

> For example, targets $(1, 0, 1, 0, 0, 1)$ and $(1, 0, 1, 1, 1, 0)$ agree on the observed points. The learner is correct on all three unseen points under the first target and wrong on all three under the second. Because C is shattered, every one of its 2^6 binary labelings is the restriction of some hypothesis in $\mathcal{H}$, so both targets are realizable within the class on this support.

IN-CLASS CHECK

Prompt: Correct this statement: "No-Free-Lunch says that ERM fails on every distribution whenever the hypothesis class is infinite." Then state the contrapositive used in the Fundamental Theorem.

Hint: Repair the quantifiers and replace raw cardinality by VC dimension.

Answer:

The corrected statement is: for every learner and each finite sample size, a sufficiently rich binary problem contains some realizable distribution on which that learner fails with nontrivial probability. An infinite class may still be learnable if its VC dimension is finite. For the theorem's necessity direction, if $\mathcal{H}$ had infinite VC dimension, the learner's sample size would admit a shattered $2m$-point hard construction; therefore a realizably PAC-learnable class must have finite VC dimension.

Standard language to carry forward

Standard term or notation	Meaning in this Tutorial	Formula or proof cue
Agnostic PAC learnability	Compete with the best population risk in $\mathcal{H}$ on every joint distribution	$R(A(S)) \leq \inf_{h\in\mathcal{H}} R(h) + \varepsilon$
Excess risk	Population risk above the best-in-class infimum	$R(h) - \inf_{g\in\mathcal{H}} R(g)$
Uniform convergence	One sample controls empirical-to-population deviation for every h simultaneously	$\sup_{h\in\mathcal{H}} \left\lvert \hat{R}_{S(h)} - R(h) \right\rvert \leq \alpha$
Agnostic ERM guarantee	Two deviation moves plus optimization error	excess $\leq 2\alpha + \eta$
Fundamental Theorem of Statistical Learning	Within the binary zero-one measurable frame, finite VC dimension exactly characterizes learnability	six equivalent properties
No-Free-Lunch theorem	Every learner has a hard realizable problem when no restriction limits unseen labels	for every learner, there exists a hard problem

Reconstruct the Tutorial 3B entry

IN-CLASS CHECK

Prompt: Reconstruct the entire argument without using notes. Your response must: (i) state the agnostic PAC quantifiers; (ii) define the uniform event; (iii) derive the approximate-ERM chain with all tolerances; (iv) state the Fundamental Theorem with its scope; and (v) explain the shattered-set, random-label, and unseen-label necessity mechanism.

Hint: Use the registered symbols $\mathcal{H}$, S, R, $\hat{R}_S$, (ε, δ), and $\mathrm{VCdim}(\mathcal{H})$ consistently.

Answer:

There exist a learner A and sample-complexity function such that for every $\varepsilon, \delta \in (0,1)$ and every joint distribution D on $\mathcal{X} \times \{0,1\}$, a sufficiently large i.i.d. sample $S = ((x_i, y_i))_{i=1}^m$ yields $R(A(S)) \le \inf_{h\in\mathcal{H}} R(h) + \varepsilon$ with probability at least $1-\delta$, over the sample and any learner randomness.

The uniform event is

$$E_\alpha = \left\{ \sup_{h\in\mathcal{H}} \left| \hat{R}_{S(h)} - R(h) \right| \le \alpha \right\}. \tag{19}$$

If h_S is an η-approximate ERM and h_ρ is within ρ of the population infimum, then on E_α,

$$R(h_S) \le \hat{R}_{S(h_S)} + \alpha \le \hat{R}_{S(h_\rho)} + \alpha + \eta \le R(h_\rho) + 2\alpha + \eta \le \inf_{h\in\mathcal{H}} R(h) + \rho + 2\alpha + \eta. \tag{20}$$

Letting $\rho \to 0$ gives excess at most $2\alpha + \eta$.

In distribution-free binary classification with zero-one loss and suitable measurability, uniform convergence, agnostic ERM success, agnostic PAC learnability, realizable PAC learnability, realizable ERM success, and finite $\mathrm{VCdim}(\mathcal{H})$ are equivalent.

Necessity follows by contrapositive. Infinite VC dimension supplies a shattered set of $2m$ points. A sample of size m leaves at least half the support unseen; random target labels on those unseen points remain unpredictable, so every learner incurs constant error on average for some realizable target and hence fails with constant probability. The remaining open job is to turn finite VC behavior, with Sauer-Shelah as input, into the simultaneous uniform event.

Exact entry interface

Exact entry item	What you must be able to reconstruct	Origin
Empirical risk minimization	compare empirical risks; allow an η optimization tolerance	Chapter 2
VC dimension	constructive lower bound plus universal upper bound	Chapter 2
Sauer-Shelah lemma	finite VC dimension gives polynomial restriction growth	Chapter 2
Agnostic PAC learnability	best-in-class excess-risk quantifiers	Tutorial 3A
Uniform convergence	one supremum event for every hypothesis	Tutorial 3A
Agnostic ERM guarantee	the $2\alpha + \eta$ comparison chain	Tutorial 3A
Fundamental Theorem	the six scoped equivalent properties	Tutorial 3A
No-Free-Lunch theorem	shattered unseen labels force necessity	Tutorial 3A
$\mathcal{H}$	the declared candidate hypothesis class	stable notation
$S = ((x_i, y_i))_{i=1}^m$	an ordered i.i.d. labeled training sample	stable notation
$R(h)$	population risk under the declared distribution and loss	stable notation
$\hat{R}_{S(h)}$	sample-average empirical risk	stable notation
(ε, δ)	accuracy tolerance and failure probability	stable notation
$\mathrm{VCdim}(\mathcal{H})$	largest shattered-set size, or infinity	stable notation

Table 6: The fourteen rows are exactly the knowledge and notation Tutorial 3B may assume.

Key Takeaway

Agnostic learning replaces perfection by competition with the best member of a declared class. Uniform convergence is the event that makes this competition safe for a data-selected ERM output, and its proof reveals the exact tolerance $2\alpha + \eta$. In the standard measurable binary zero-one setting, finite VC dimension is exactly the learnability boundary. Infinite VC dimension fails because a finite sample cannot reveal random labels on a large shattered set. The open finite-VC-to-uniform arrow is where Tutorial 3B begins; a technique called symmetrization will be defined there, not used here.

Optional self-study extension

OPTIONAL DEEP DIVE - 0 CLASS MINUTES

Question: How do the random-label and unseen-point ideas yield the standard No-Free-Lunch constants $\frac{1}{8}$ and $\frac{1}{7}$?

Answer.

Fix a learner A and a sample size m. Choose a set C of $2m$ points. In the class-level necessity argument, C is shattered, so all target labelings below are realized by hypotheses in $\mathcal{H}$. For each binary function $f : C \to \{0, 1\}$, let D_f be uniform on the labeled pairs $\{(x, f(x)) : x \in C\}$. Then $R_{D_f}(f) = 0$.

Average uniformly over all 2^{2m} functions f, over $S \sim D_f^m$, and over any randomness of A. After conditioning on the sampled inputs and observed labels, at least m support points remain unseen. Their target bits are conditionally independent and uniform, so every learner prediction there is wrong with probability $\frac{1}{2}$. The unseen mass is at least $\frac{m}{2m} = \frac{1}{2}$; hence conditional expected risk is at least $\left(\frac{1}{2}\right)\left(\frac{1}{2}\right) = \frac{1}{4}$.

Averaging cannot exceed the largest component. Hence there is at least one fixed target $f^\star$ for which

$$\mathbb{E}_{S \sim D_{f^\star}^m, A}\left[R_{D_{f^\star}}(A(S))\right] \geq \frac{1}{4}. \tag{21}$$

Let $Z = R_{D_{f^\star}}(A(S))$, so $0 \leq Z \leq 1$, and write $p = \mathbb{P}\left(Z \geq \frac{1}{8}\right)$. Splitting the expectation at $\frac{1}{8}$ gives

$$\mathbb{E}[Z] \leq (1-p)\left(\frac{1}{8}\right) + p(1) = \frac{1}{8} + 7\frac{p}{8}. \tag{22}$$

Combining this with $\mathbb{E}[Z] \geq \frac{1}{4}$ gives $p \geq \frac{1}{7}$. Thus the learner has risk at least $\frac{1}{8}$ with probability at least $\frac{1}{7}$, although the fixed target $f^\star$ has risk zero. This completes the averaging, occupancy, and constant-probability bookkeeping.

Post-class or self-study. Not a prerequisite for the core path.

Source notes

The official Week 3 Learning Sheet v4 is used only to preserve the assigned scope and sequence: `w03-agnostic-pac`, printed/PDF pp. 1-6; `w03-uniform-convergence`, pp. 7-16; and `w03-fundamental-theorem`, pp. 17-27. The later random-sign, ghost-sample, symmetrization, rate, and model-selection units are not taught here. Their proof machinery belongs to Tutorial 3B.

Definitions, results, arguments, and examples were reconstructed from:

- Shalev-Shwartz and Ben-David, *Understanding Machine Learning*, Section 3.2.1 and Definition 3.3 (printed/PDF pp. 44-47), for the joint data-label distribution and agnostic best-in-class guarantee;
- Shalev-Shwartz and Ben-David, Sections 4.1-4.2, especially Lemma 4.1, Definition 4.3, Corollaries 4.4 and 4.6 (printed/PDF pp. 54-58), for uniform convergence, the ERM comparison, and the finite-class calculation;
- Shalev-Shwartz and Ben-David, Theorem 5.1 and Section 5.1 (printed/PDF pp. 61-64), for No-Free-Lunch, unseen labels, and the optional constant bookkeeping;
- Shalev-Shwartz and Ben-David, Theorems 6.7-6.8 and Sections 6.4-6.5 (printed/PDF pp. 72-78), for the Fundamental Theorem, implication map, and finite-VC proof boundary;
- Mohri, Rostamizadeh, and Talwalkar, *Foundations of Machine Learning*, 2nd ed., Section 3.1 (printed pp. 30-34; PDF pp. 47-51), for the convention that the relevant suprema and risk events are measurable.

The full No-Free-Lunch bookkeeping is optional, receives zero class minutes, and exports no knowledge required by the core path. All required teaching uses only binary zero-one classification and the stated measurability boundary.

DSAA3073: Theories in Data Science

Tutorial 3B: Build the Missing Generalization Bound

150-minute core tutorial - 15-minute break - 60-minute protected buffer

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Explorative projection

LESSON BACKBONE

Finite VC dimension limits label patterns. How can an independent sample and random signs turn that combinatorial fact into a uniform statistical guarantee, and then into a principled way to choose a model class?

Tutorial 3A left one dashed arrow open: finite VC dimension should imply uniform convergence. Sauer-Shelah controls how many restrictions a binary class can realize on a fixed finite set, but the deviation we need contains a population expectation and a class selected through the random sample.

We will first measure how strongly a loss class can align with random signs. Then an independent copy of the sample will convert the population deviation into a paired empirical process, and exchangeability will expose exactly two random-sign terms. The final block reads the resulting sample rates, turns simultaneous class bounds into structural risk minimization, and makes the choice of proof method explicit.

The three core blocks take **35, 70, and 45 minutes**. Take the scheduled 15-minute break after the first block. The last 60 minutes are protected for questions, reconstruction, and recovery; they introduce no required knowledge.

Clock	Mathematical work	Protected time
0-35	Define empirical Rademacher complexity and calculate it exactly	35 min core
35-50	Scheduled break	15 min
50-120	Build the ghost-sample and symmetrization proof chain	70 min core
120-165	Compare rates, derive SRM, and select a proof route	45 min core
165-225	Questions, recovery, alternative derivations	60 min buffer

Let noise probe the loss class

Tutorial 3A ended with fourteen entry items. The mathematical ones we use immediately are the hypothesis class $\mathcal{H}$, an i.i.d. labeled sample $S = ((x_i, y_i))_{i=1}^{m}$, population risk $R(h)$, empirical risk $\hat{R}_{S(h)}$, finite $\mathrm{VCdim}(\mathcal{H}) = d$, and the Sauer-Shelah restriction bound. The agnostic ERM comparison tells us why the target is the uniform event

$$\sup_{h\in\mathcal{H}} \left| R(h) - \hat{R}_{S(h)} \right| \le \alpha. \tag{23}$$

The Fundamental Theorem says that finite d is sufficient. It did not yet give the statistical proof.

Write $\mathcal{Z} = \mathcal{X} \times \{0, 1\}$ and associate each hypothesis with its zero-one loss function

$$g_{h(z)} = \mathbf{1}_{\{h(x)\neq y\}}, \quad z = (x, y), \quad \mathcal{G} = \{g_h : h \in \mathcal{H}\}. \tag{24}$$

The distinction matters: $\mathcal{H}$ contains predictors $x \mapsto h(x)$, while $\mathcal{G}$ contains real-valued loss functions $z \mapsto g_{h(z)}$. On the fixed sample, each g produces a vector $g_S = (g(z_1), ..., g(z_m))$.

Suppose a fresh vector of independent signs $\sigma = (\sigma_1, ..., \sigma_m)$ is drawn, with $\mathbb{P}(\sigma_i = +1) = \mathbb{P}(\sigma_i = -1) = \frac{1}{2}$. The sample stays fixed. For every sign vector, ask which loss vector has the largest normalized inner product with σ.

$\boldsymbol{\sigma_1}$	$\boldsymbol{\sigma_2}$	$\boldsymbol{\sigma_3}$	$\boldsymbol{\sigma \cdot \frac{g_a}{3}}$	$\boldsymbol{\sigma \cdot \frac{g_b}{3}}$
$+1$	$+1$	$+1$	0	$\frac{2}{3}$
$+1$	$+1$	-1	0	$\frac{2}{3}$
$+1$	-1	$+1$	0	0
$+1$	-1	-1	0	0
-1	$+1$	$+1$	0	0
-1	$+1$	-1	0	0
-1	-1	$+1$	0	$-\frac{2}{3}$
-1	-1	-1	0	$-\frac{2}{3}$

Table 7: On this fixed three-example sample, $g_a = (0, 0, 0)$ and $g_b = (1, 1, 0)$. For each sign row, maximize across the two loss columns before averaging across rows.

The maximizing value is $\frac{2}{3}$ in the first two rows and 0 in the other six, because g_a is always available. Its average is $\frac{[2(\frac{2}{3})+6(0)]}{8} = \frac{1}{6}$. The order is visible: fix S, draw σ, maximize over $\mathcal{G}$, then average over σ.

Definition: Empirical Rademacher complexity

Let $\mathcal{G}$ be a class of real-valued functions on $\mathcal{Z}$, and let $S = (z_1, ..., z_m)$ be fixed. Its **empirical Rademacher complexity** is

$$\hat{\mathfrak{R}}_{S(\mathcal{G})} = \mathbb{E}_\sigma \left[\sup_{g \in \mathcal{G}} \frac{1}{m} \sum_{i=1}^{m} \sigma_i g(z_i) \right], \tag{25}$$

where the σ_i are independent and uniform on $\{-1, +1\}$, independent of the sample. We assume the displayed supremum is measurable.

The registered symbol is $\hat{\mathfrak{R}}_{S(\mathcal{G})}$. We do not use a script R for this quantity because $R(h)$ is already population risk. The distribution-averaged version is

$$\mathfrak{R}_{m(\mathcal{G})} = \mathbb{E}_{S \sim D^m} \left[\hat{\mathfrak{R}}_{S(\mathcal{G})} \right]. \tag{26}$$

The empirical quantity conditions on the observed sample; the expected quantity also averages over samples. In the table above, $\hat{\mathfrak{R}}_{S(\mathcal{G})} = \frac{1}{6}$.

Worked example: A class that can fit every sign

Suppose the restriction vectors of $\mathcal{G}$ contain every vector in $\{-1, +1\}^m$. For each sign vector σ, choose g with $g_S = \sigma$. Then

$$\sup_{g \in \mathcal{G}} \frac{1}{m} \sigma \cdot g_S = 1, \tag{27}$$

so $\hat{\mathfrak{R}}_{S(\mathcal{G})} = 1$. A class with fewer available restriction vectors cannot necessarily follow every random sign. The quantity therefore measures richness on the observed inputs, not training accuracy on the observed labels.

IN-CLASS CHECK

Prompt: For the eight-row table, explain why averaging each loss vector over signs before maximizing gives the wrong answer. Then recompute the correct value.

Hint: The maximizing loss function may depend on the realized sign vector.

Answer:

If we average first, both fixed signed sums have expectation zero and their maximum is zero. The definition instead takes $\mathbb{E}_{\sigma[\sup_g]}$: it may choose g_b in the two positive-positive rows and g_a elsewhere. Hence the correct value is $\frac{\frac{2}{3}+\frac{2}{3}}{8} = \frac{1}{6}$. In general, $\sup_g \mathbb{E}_{\sigma[X_g]}$ need not equal $\mathbb{E}_{\sigma[\sup_g X_g]}$.

IN-CLASS CHECK

Prompt: Classify each object as fixed, random, or optimized in $\hat{\mathfrak{R}}_{S(\mathcal{G})}$: S, σ, g, and $\mathcal{G}$. What extra randomness enters $\mathfrak{R}_{m(\mathcal{G})}$?

Hint: Read the operations from the inside outward.

Answer:

> For the empirical quantity, S and the declared class $\mathcal{G}$ are fixed, σ is random, and g is optimized separately for each realized σ. The expected complexity adds an outer expectation over the random sample $S \sim \mathcal{D}^m$.

An unknown $R(h)$ does not invalidate probability analysis. Hoeffding's inequality routinely bounds deviations from unknown expectations, and a union bound is fully valid for a fixed finite $\mathcal{H}$. The obstacle for an infinite class is subtler. The set of restrictions $\mathcal{H}|_S$ depends on the same random sample, and two hypotheses that agree on S can have the same empirical risk but different population risks. Counting only their observed restrictions therefore does not directly control the population-deviation events.

We need a finite **paired empirical process** whose behavior can be counted after conditioning on its sampled points. That is the job of an independent copy of S.

Replace the population by an independent empirical copy

Before naming the proof devices, inspect the factor ledger. Let $\hat{E}_{S[g]} = m^{-1}\sum_i g(z_i)$. The four stages are

stage	process	factor entering	reason for next arrow
1	$\mathbb{E}_S \sup_{g(\mathbb{E}_D g-\hat{E}_S g)}$	-	independent empirical copy
2	$\mathbb{E}_{S,S'} \sup_{g(\hat{E}_{S'} g-\hat{E}_S g)}$	1	$\sup \mathbb{E} \le \mathbb{E} \sup$
3	$\mathbb{E}_{S,S',\sigma} \sup_g m^{-1} \sum_i \sigma_{i(g(z_{i'})-g(z_i))}$	1	pair swapping preserves the law
4	$2\mathfrak{R}_{m(\mathcal{G})}$	2	supremum subadditivity

σ_i	subtracted slot		added slot	observable action and answer
$+1$	z_i	$\to$	$z_{i'}$	keep $(z_i, z_{i'})$; contribution $g(z_{i'}) - g(z_i)$
-1	$z_{i'}$	$\to$	z_i	swap to $(z_{i'}, z_i)$; contribution $g(z_i) - g(z_{i'})$

Table 8: Cover the last column, use the sign to place each paired row, then reveal the answer. The ghost step still has factor one: $\sigma_i = +1$ keeps the orientation and $\sigma_i = -1$ swaps it. Exchangeability licenses the later signed equality; only the final supremum split creates factor two.

Because each pair is i.i.d., the keep and swap orientations have the same distribution. The switch is the concrete mechanism behind both definitions that follow.

Definition: Ghost sample

Given $S = (z_1, \ldots, z_m) \sim D^m$, a **ghost sample** $S' = (z_{1'}, \ldots, z_{m'}) \sim D^m$ is an independent i.i.d. sample of the same size from the same distribution. It is introduced only inside the proof so that $\mathbb{E}_{S'}\left[\hat{E}_{S'}[g]\right] = \mathbb{E}_{D[g]}$ for every fixed g.

The algorithm never receives S'. It is not a validation set, a second training set, or extra data promised to the learner. Independence lets the analysis replace a population expectation by an expectation of an empirical average without changing the algorithm.

IN-CLASS CHECK

Prompt: A practitioner splits the observed data into training and validation sets and reports validation error. Is the validation set the ghost sample in this proof? State one similarity and two differences.

Hint: Separate a proof distribution from data actually consumed by a procedure.

Answer:

> Both are intended to be independent of the training data and drawn from the same population. The ghost sample, however, is integrated out inside the proof and is not observed by the algorithm; a validation set is real data used by a procedure or evaluation. The proof also fixes the ghost sample size to match S so that observations can be paired, whereas a validation split need not have that size.

Definition: Rademacher symmetrization

Rademacher symmetrization bounds a worst-case population-minus-empirical deviation by a paired empirical difference, uses exchangeability of the sample pairs to insert independent random signs, and splits the paired supremum into two Rademacher-complexity terms.

We now justify every arrow. Let $\mathcal{G}$ map $\mathcal{Z}$ into $[0,1]$, assume the relevant suprema are measurable, and define

$$\Phi(S) = \sup_{g\in\mathcal{G}}\Big(\mathbb{E}_{D[g]} - \hat{E}_{S[g]}\Big). \tag{28}$$

For every fixed S,

$$\begin{aligned}\Phi(S) &= \sup_g \mathbb{E}_{S'}\left[\hat{E}_{S'}[g] - \hat{E}_{S[g]}\right] \\ &\le \mathbb{E}_{S'}\left[\sup_{g\left(\hat{E}_{S'}[g] - \hat{E}_{S[g]}\right)}\right].\end{aligned} \tag{29}$$

The inequality is $\sup_g \mathbb{E}\left[X_g\right] \le \mathbb{E}\left[\sup_g X_g\right]$. Taking expectation over S gives

$$\mathbb{E}_{S[\Phi(S)]} \le \mathbb{E}_{S,S'}\left[\sup_g \frac{1}{m}\sum_{i=1}^{m(g(z_{i'})-g(z_i))}\right]. \tag{30}$$

There is no factor two here.

Now draw independent Rademacher signs. For each i, multiplying $g(z_{i'}) - g(z_i)$ by $\sigma_i = -1$ is the same as swapping z_i and $z_{i'}$. The joint law is unchanged, so

$$\begin{aligned}&\mathbb{E}_{S,S'}\left[\sup_g \frac{1}{m}\sum_{i(g(z_{i'})-g(z_i))}\right] \\ &= \mathbb{E}_{S,S',\sigma}\left[\sup_g \frac{1}{m}\sum_i \sigma_{i(g(z_{i'})-g(z_i))}\right].\end{aligned} \tag{31}$$

Finally, subadditivity of the supremum gives

$$\begin{aligned}&\mathbb{E}_{S,S',\sigma}\left[\sup_g \frac{1}{m}\sum_i \sigma_{i(g(z_{i'})-g(z_i))}\right] \\ &\le \mathbb{E}_{S',\sigma}\left[\sup_g \frac{1}{m}\sum_i \sigma_i g(z_{i'})\right] + \mathbb{E}_{S,\sigma}\left[\sup_g \frac{1}{m}\sum_i (-\sigma_i) g(z_i)\right] \\ &= 2\mathfrak{R}_{m(\mathcal{G})}.\end{aligned} \tag{32}$$

The last equality uses the identical distributions of S and S' and of σ and $-\sigma$. This is the only step in the chain that creates the factor two.

Result: Expected one-sided symmetrization bound

For a measurable class $\mathcal{G}$ of functions $\mathcal{Z} \to [0,1]$,

$$\mathbb{E}_{S\left[\sup_{g\in\mathcal{G}}\left(\mathbb{E}_{D[g]}-\hat{E}_{S[g]}\right)\right]} \le 2\mathfrak{R}_{m(\mathcal{G})}. \tag{33}$$

The ghost-sample step has coefficient one; the random-sign split produces the final coefficient two.

IN-CLASS CHECK

Prompt: Write 1, 1, or 2 under the three arrows: population deviation to ghost difference; ghost difference to signed paired process; signed paired process to separate Rademacher terms. Name the justification for each.

Hint: The first two transformations do not split a supremum into two pieces.

Answer:

> The ledger is $1, 1, 2$. The first arrow uses $\sup_g \mathbb{E}[X_g] \le \mathbb{E}[\sup_g X_g]$; the second uses exchangeability of every i.i.d. pair under sign-controlled swapping; the third uses $\sup_g(A_g+B_g) \le \sup_g A_g + \sup_g B_g$, producing two identically distributed Rademacher terms.

IN-CLASS CHECK

Prompt: For one pair $(z_i, z_{i'})$, list the summand when $\sigma_i = +1$ and when $\sigma_i = -1$. Why is averaging over these signs legitimate only because the pair is exchangeable?

Hint: A negative sign reverses the difference.

Answer:

> The summands are $g(z_{i'}) - g(z_i)$ and $g(z_i) - g(z_{i'})$. The second is obtained by swapping the pair. Because z_i and $z_{i'}$ are independent draws from the same distribution, the ordered pairs $(z_i, z_{i'})$ and $(z_{i'}, z_i)$ have the same law. Without identical distributions or with dependence that breaks exchangeability, inserting the sign could change the expectation.

From the expected process to a probability statement

Changing one point of a $[0,1]$-valued sample changes $\Phi(S)$ by at most $\frac{1}{m}$. Bounded-difference concentration therefore upgrades the expectation bound. One form of Mohri, Rostamizadeh, and Talwalkar's Theorem 3.3 is:

Result: Rademacher generalization bound

For any $\delta \in (0,1)$, with probability at least $1-\delta$ over $S \sim D^m$, every $g \in \mathcal{G}$ satisfies

$$\mathbb{E}_{D[g]} \le \hat{E}_{S[g]} + 2\mathfrak{R}_{m(\mathcal{G})} + \sqrt{\frac{\ln(\frac{1}{\delta})}{2m}}. \tag{34}$$

A data-dependent version replaces the expected complexity by the observed $\hat{\mathfrak{R}}_{S(\mathcal{G})}$ and has the valid bound

$$\mathbb{E}_{D[g]} \le \hat{E}_{S[g]} + 2\hat{\mathfrak{R}}_{S(\mathcal{G})} + 3\sqrt{\frac{\ln(\frac{2}{\delta})}{2m}}. \tag{35}$$

Apply the same reasoning to $-\mathcal{G}$, or split the confidence budget between the two directions, to control absolute deviations. The exact constants depend on the chosen one-sided or two-sided form; the structural message is stable: a Rademacher term controls the class-wide search, and a concentration term controls the remaining sample fluctuation.

IN-CLASS CHECK

Prompt: Correct the statement: "A direct probability bound is impossible because $R(h)$ is unknown, so the ghost sample makes it computable." What is the actual obstacle, and what does the ghost sample create?

Hint: Unknown expectations are normal in concentration inequalities. Focus on sample-dependent restriction classes.

Answer:

The numerical value of $R(h)$ need not be known for Hoeffding or another probability inequality to be valid. The obstacle is that an infinite class reduced to its observed restrictions is itself sample-dependent, and equality on S does not control population deviations. The independent ghost sample creates a paired empirical process on sampled points. After conditioning on those points, its finite restriction behavior can be controlled combinatorially or through Rademacher complexity.

Close the dashed finite-VC arrow

For a binary class written with values in $\{-1,+1\}$, the restriction set on $S_X = (x_1, \dots, x_m)$ is

$$V_S = \{(h(x_1), \dots, h(x_m)) : h \in \mathcal{H}\} \subseteq \{-1,+1\}^m. \tag{36}$$

Every vector in V_S has Euclidean norm $\sqrt{m}$. Massart's lemma therefore gives

$$\hat{\mathfrak{R}}_{S_X}(\mathcal{H}) \le \sqrt{2\ln\frac{|V_S|}{m}} \le \sqrt{2\ln\Pi_{\mathcal{H}}\frac{m}{m}}. \tag{37}$$

If $\mathrm{VCdim}(\mathcal{H}) = d$ and $m \ge d$, Sauer-Shelah supplies

$$\Pi_{\mathcal{H}}(m) \le \sum_{j=0}^{d}\binom{m}{j} \le \left(e\frac{m}{d}\right)^d. \tag{38}$$

For zero-one classification, the loss class $\mathcal{G}$ is obtained from $\mathcal{H}$ and the labels; the corresponding Rademacher and growth bounds yield a two-sided uniform-deviation scale

$$\sup_{h\in\mathcal{H}}\left|R(h)-\hat{R}_{S(h)}\right| = O\left(\sqrt{\frac{d\ln\left(e\frac{m}{d}\right)+\ln\left(\frac{1}{\delta}\right)}{m}}\right) \tag{39}$$

with probability at least $1-\delta$. The logarithmic form shown here is a standard elementary closure through Massart plus Sauer. Sharper analyses can remove or reorganize some logarithmic factors, but they do not alter the finite-VC learnability conclusion.

This completes Tutorial 3A's dashed arrow within the stated binary zero-one, distribution-free i.i.d. setting. As before, we require suitable measurability of the risks, suprema, and events. The caveat rules out pathological formal objects; it is not a source of statistical complexity and not a step that changes the bound.

IN-CLASS CHECK

Prompt: Put these four cards in causal order: Sauer-Shelah; Massart's lemma; Rademacher generalization bound; finite-VC uniform convergence. State what each card contributes.

Hint: Move from a combinatorial count to a random-sign average and then to a population deviation.

Answer:

> Sauer-Shelah bounds the number of restriction vectors by polynomial growth. Massart converts the logarithm of that finite count into a Rademacher bound. The symmetrization and concentration theorem converts Rademacher complexity into a high-probability uniform population-versus-empirical bound. Since the resulting radius tends to zero for fixed finite d, uniform convergence follows.

Worked example: Read a VC-controlled radius

Ignore universal constants and take the displayed order bound with $d=10$, $m=1000$, and $\delta=0.05$. Its numerator is

$$10\ln\left(\frac{e(1000)}{10}\right)+\ln(20)\approx 59.05. \tag{40}$$

The resulting scale is $\sqrt{\frac{59.05}{1000}}\approx 0.243$. This is a distribution-free worst-case scale, not a calibrated numerical confidence interval. The calculation is useful for comparing how d, m, and δ enter, while theorem-specific constants must be retained for an actual certificate.

Read the rate before choosing the class

The Fundamental Theorem has two quantitative regimes. Hold the VC dimension d and confidence δ fixed, then ask what happens when the target ε is halved.

regime	representative sample scale	leading epsilon power	halving ε
realizable PAC	$\Omega\left(\frac{d+\ln(\frac{1}{\delta})}{\varepsilon}\right)$ to $O\left(\frac{d\ln(\frac{1}{\varepsilon})+\ln(\frac{1}{\delta})}{\varepsilon}\right)$	$\frac{1}{\varepsilon}$	about $2\times$, plus logs
agnostic PAC / uniform convergence	$\Theta\left(\frac{d+\ln(\frac{1}{\delta})}{\varepsilon^2}\right)$ up to absolute constants	$\frac{1}{\varepsilon^2}$	about $4\times$

Table 9: Realizable learning detects consistency; agnostic learning estimates an excess-risk comparison in noise. The upper and lower statements must keep their logarithmic qualifications.

Definition: VC sample-complexity rates

For binary zero-one classes with finite VC dimension d, realizable PAC learning has leading accuracy dependence $\frac{1}{\varepsilon}$, with a standard upper bound containing $d\ln(\frac{1}{\varepsilon})$ and a lower bound of order $\frac{d+\ln(\frac{1}{\delta})}{\varepsilon}$. Agnostic PAC learning and uniform convergence have leading dependence $\frac{1}{\varepsilon^2}$, with matching $\frac{d+\ln(\frac{1}{\delta})}{\varepsilon^2}$ upper and lower orders up to absolute constants.

The difference is not a contradiction. In the realizable case, a consistent hypothesis can be rejected when it makes a population error on a sampled point. In the agnostic case, the learner must estimate a difference between nonzero risks accurately enough to compete with the best member of the class; mean estimation naturally produces the squared accuracy dependence.

IN-CLASS CHECK

Prompt: A noisy classification problem requires excess risk at most ε. A proposed sample calculation scales like $\frac{1}{\varepsilon}$. Which assumption would be needed to justify that leading rate, and why is it unavailable here?

Hint: Distinguish consistency with a perfect target from estimation around a positive optimum.

Answer:

> The leading $\frac{1}{\varepsilon}$ route requires the realizability assumption: some member of the class has zero population loss and a consistent output can be analyzed through the probability of missing its error region. In a noisy agnostic problem the best class risk may be positive, so success requires estimating risk differences. The distribution-free agnostic rate has leading $\frac{1}{\varepsilon^2}$ dependence.

Let every class carry its own certificate

Suppose the search space is a countable union $\mathcal{H} = \cup_{k\in\mathbb{N}} \mathcal{H}_k$, and every $\mathcal{H}_k$ has a valid uniform deviation radius $\varepsilon_{k(m,\delta_k)}$ for $\delta_k \in (0,1)$. For every active class choose $w_k > 0$ with $\sum_k w_k = 1$, and set $\delta_k = w_k\delta$. For each fixed k, the class bound fails with probability at most δ_k; a union bound gives total failure probability at most $\sum_k \delta_k = \delta$. Thus, with probability at least $1-\delta$, simultaneously for every k and $h \in \mathcal{H}_k$,

$$R(h) \leq \hat{R}_{S(h)} + \varepsilon_{k(m,w_k\delta)}. \tag{41}$$

nested class	w_k	$\widehat{R}_S$	penalty	bound	certificate route
$\mathcal{H}_1$	$\frac{1}{2}$	0.18	0.06	0.24	finite union
$\mathcal{H}_2 \supset \mathcal{H}_1$	$\frac{1}{3}$	0.11	0.10	**0.21**	VC growth
$\mathcal{H}_3 \supset \mathcal{H}_2$	$\frac{1}{6}$	0.08	0.16	0.24	Rademacher

Table 10: The weights sum to one. Training error favors the largest nested class, but the smallest certified simultaneous upper bound selects $\mathcal{H}_2$. Each penalty names the proof route that supports it.

Definition: Structural risk minimization

Structural risk minimization (SRM) chooses across a declared finite or countable family of classes by minimizing a simultaneous class-specific upper bound,

$$\hat{R}_{S(h)} + \varepsilon_{k(m,w_k\delta)}, \quad h \in \mathcal{H}_k, \tag{42}$$

where every active class has $w_k > 0$, $\sum_k w_k = 1$, and each penalty comes from a justified uniform bound for its class. A zero-weight class is excluded from the active family (equivalently, it has infinite penalty). If a hypothesis lies in several active classes, use the smallest valid bound assigned to it.

This is more than adding an attractive square-root term to training error. The class family, confidence allocation, and theorem supporting every ε_k must be declared. Without them, the expression may be a useful heuristic but is not the stated distribution-free SRM guarantee.

IN-CLASS CHECK

Prompt: Using the table, select the SRM class and explain why empirical risk alone gives a different answer. If the weights summed to 1.2, would the confidence argument still certify failure probability at most δ?

Hint: Add the last two columns and inspect the union-bound budget.

Answer:

> SRM selects $\mathcal{H}_2$ because $0.11 + 0.10 = 0.21$ is smaller than both 0.24 bounds. Empirical risk alone would select $\mathcal{H}_3$ at 0.08 and ignore its larger estimation uncertainty. If the weights summed to 1.2, the union bound would give failure probability at most 1.2δ, not δ; the advertised confidence certificate would fail.

Match the proof to the available structure

The same target statement can admit several valid bounds. The shortest sound route depends on what is known about the class and whether the goal is an upper bound or an impossibility result.

available structure	**target**	**first route to try**	**decisive object**
fixed finite $\mathcal{H}$	uniform upper bound	Hoeffding plus union bound	$\ln\lvert\mathcal{H}\rvert$
binary class, finite VC dimension	distribution-free upper bound	Sauer/growth plus Massart	$\Pi_{\mathcal{H}}(m)$ and d
real-valued loss class or useful data-dependent geometry	generalization upper bound	ghost sample plus Rademacher symmetrization	$\hat{\mathfrak{R}}_{S(\mathcal{G})}$
claimed universal learner or rate is too strong	lower bound / impossibility	shattered set and indistinguishable labels	unseen alternatives

Table 11: Infinite cardinality alone does not force one route. Use the simplest structure that actually controls the target statement.

Course strategy - proof-route decision aid

This course uses the table as a decision procedure: match the target claim and available class structure to a finite union bound, growth/VC control, Rademacher symmetrization, or a shattered-set lower-bound construction. This is a course strategy for organizing known tools, not a standard named textbook definition.

IN-CLASS CHECK

Prompt: Choose a route for each claim: (i) a library of 500 fixed classifiers; (ii) binary halfspaces with known finite VC dimension; (iii) a Lipschitz real-valued loss class with a directly bounded empirical random-sign complexity; (iv) a claim that every binary labeling rule is learnable from m samples.

Hint: Ask whether the claim is an upper or lower bound, then identify the strongest available structure.

Answer:

(i) Use fixed-hypothesis concentration and a union bound over 500 rules. (ii) Use the VC-growth route, with symmetrization providing the statistical bridge when needed. (iii) Use Rademacher symmetrization directly because its empirical complexity captures the real-valued loss geometry. (iv) Use a shattered-set or No-Free-Lunch lower-bound construction: the target is an impossibility statement, so an upper-bound route cannot settle it.

Standard language to carry forward

Standard term or notation	Meaning in this Tutorial	Formula or proof cue
Empirical Rademacher complexity	Random-sign fit of a function class on a fixed sample	$\hat{\mathfrak{R}}_{S(\mathcal{G})} = \mathbb{E}_\sigma \sup_g m^{-1} \sum_i \sigma_i g(z_i)$
Ghost sample	Independent same-distribution sample used only inside the proof	$S' \sim D^m$, independent of S
Rademacher symmetrization	Population deviation to paired process to two signed suprema	factor ledger $1, 1, 2$
VC sample-complexity rates	Realizable and agnostic regimes have different leading accuracy powers	$\frac{1}{\varepsilon}$ versus $\frac{1}{\varepsilon^2}$
Structural risk minimization	Minimize a simultaneous empirical-risk-plus-penalty certificate over active classes	$\delta_k = w_k\delta,\quad w_k > 0,\ \sum_k w_k = 1$

Reconstruct the chapter boundary

The Chapter 4 entry has exactly three items. Two are earlier mathematical background, and one is new here.

IN-CLASS CHECK

Prompt: Without notes, reconstruct: (i) the definition of a Hilbert space at the level established in Chapter 1; (ii) the linear-algebra facts that let matrices and quadratic forms be read spectrally; and (iii) the SRM principle. Then state the common question these three views create for the next chapter.

Hint: Keep the first two items as recalled prerequisites. Do not introduce a new regression formula.

Answer:

A Hilbert space is a complete inner-product space, with norm induced by $\|f\| = \sqrt{\langle f, f\rangle}$; completeness makes limits of Cauchy sequences remain in the space. Linear algebra supplies matrices, eigenvalues, orthogonal directions, and the fact that a quadratic form or linear map can behave differently along different spectral directions. SRM chooses a hypothesis by minimizing empirical risk plus a class-specific simultaneous confidence penalty, with strictly positive active-class weights summing to one; a zero-weight class is excluded from the active family.

Together they ask how a continuous estimator's geometry and spectral directions determine the tradeoff between fitting the observed sample and controlling instability or complexity. Chapter 4 will formalize that question through bias-variance analysis and regularization; those objects are previews, not tools used in this Tutorial.

Key Takeaway

Fixing the sample before drawing random signs makes $\hat{\mathfrak{R}}_{S(\mathcal{G})}$ a precise measure of how well a loss class can follow noise. An independent ghost sample turns the population deviation into a paired empirical process. Exchangeability inserts the signs without changing the coefficient, and supremum subadditivity creates exactly two Rademacher terms. Massart plus Sauer then closes the finite-VC-to-uniform arrow. The resulting rates distinguish realizable detection from agnostic estimation, while SRM uses valid simultaneous class penalties to make model choice part of the guarantee rather than an afterthought.

Source notes

The official Week 3 Learning Sheet v4 supplies the assigned sequence and scope for `w03-rademacher` (printed/PDF pp. 28-29), `w03-symmetrization` (pp. 30-41), and `w03-lower-bounds-srm` (pp. 42-49). Two historical claims in that sheet are not retained. An unknown numerical value of $R(h)$ is not what invalidates probability analysis; the relevant obstacle is the sample-dependent restriction class and its population deviations. Also, its equation (42) places an extra factor two before the ghost-sample difference. The corrected chain above follows the original textbook proof: the ghost step has no multiplicative two, and the factor appears only after the signed paired supremum is split.

Definitions, results, and proof steps were reconstructed from:

- Mohri, Rostamizadeh, and Talwalkar, *Foundations of Machine Learning*, 2nd ed., Definitions 3.1-3.2 and Theorem 3.3 (printed pp. 30-33; PDF pp. 47-50), for empirical and expected Rademacher complexity, the ghost-sample inequality, exchangeability, the two-term split, and the high-probability bounds;
- Mohri et al., Massart's lemma and Corollaries 3.8-3.9 (printed pp. 35-36; PDF pp. 52-53), together with the previously established Sauer-Shelah lemma, for the growth-function and VC closure;
- Shalev-Shwartz and Ben-David, *Understanding Machine Learning*, Theorem 6.8 (printed/PDF pp. 72-73), for the realizable and agnostic sample-complexity regimes and their logarithmic qualifications;
- Shalev-Shwartz and Ben-David, Section 7.2, especially Theorem 7.4 (printed/PDF pp. 85-88), for summable confidence allocation and structural risk minimization.

All core statements remain within bounded measurable function classes; the VC conclusion is used for distribution-free binary zero-one classification. No optional material carries a prerequisite, and no required activity uses the protected 60-minute buffer.